



EAST WEST UNIVERSITY

Course Details

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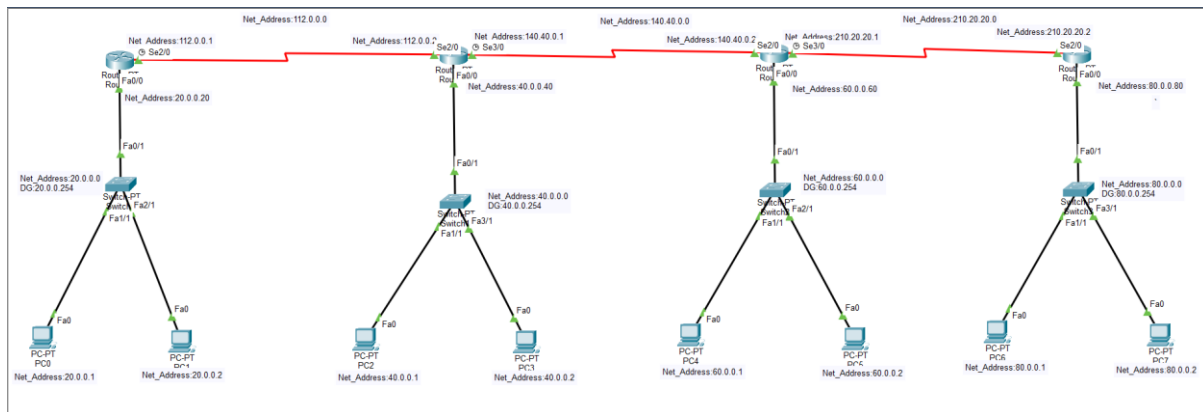


Figure: A network design with four routers

It represents a Wide Area Network (WAN) setup that connects four different Local Area Networks (LANs) using four routers. Each LAN contains two PCs connected through a switch and all routers are connected in a serial chain to form an internetwork.

The main purpose of this network design is to establish communication among all four LANs by using unique IP addressing, proper gateway configuration and router to router serial connections. This kind of topology is commonly used in large organizations where different branches are in separate areas but are connected through WAN links.

Network Components

It consists of the following main components:

Device Type	Quantity	Purpose
Routers	4	Connect different LANs and establish WAN communication
Switches	4	Connect PCs within a LAN
PCs	8	End-user devices for data communication
Serial DCE	3	Used for router-to-router WAN connections
FastEthernet Cables	12	Used for connecting switches to PCs and routers

● Network Design and IP Addressing

Each LAN and WAN connection is assigned a unique network address to ensure there are no IP conflicts.

LAN IP Plan

LAN	Router Interface	Network Address	Default Gateway (Router LAN Port)	PC Addresses
LAN 1	Router1 – Fa0/0	20.0.0.0	20.0.0.254	PC0 → 20.0.0.1 PC1 → 20.0.0.2
LAN 2	Router2 – Fa0/0	40.0.0.0	40.0.0.254	PC2 → 40.0.0.1 PC3 → 40.0.0.2
LAN 3	Router3 – Fa0/0	60.0.0.0	60.0.0.254	PC4 → 60.0.0.1 PC5 → 60.0.0.2
LAN 4	Router4 – Fa0/0	80.0.0.0	80.0.0.254	PC6 → 80.0.0.1 PC7 → 80.0.0.2

● WAN IP Plan

Link Between Routers	Router 1 st IP	Router 2 nd IP	Network Address
Router1 ↔ Router2	112.0.0.1	112.0.0.2	112.0.0.0
Router2 ↔ Router3	140.40.0.1	140.40.0.2	140.40.0.0
Router3 ↔ Router4	210.20.20.1	210.20.20.2	210.20.20.0

Working Principle

1. Each LAN has its own unique IP network connected to a router.
2. Routers use serial DCE connections to form a WAN backbone.
3. Switches provide internal LAN communication between PCs.
4. When a PC sends data to another PC in a different LAN:
 - The data goes to its default gateway (router).

- The router checks its routing table to find the destination network.
- The packet is then forwarded across serial DCE through intermediate routers until it reaches the destination router.
- Finally, the destination router delivers the data to the target PC within its LAN.

After successful configuration, all PCs can communicate each other across all LANs.

Conclusion

It successfully demonstrates how multiple LANs can be connected using routers and serial links to create a wide area network.

The setup includes proper IP addressing, gateway assignment, and routing configuration, ensuring full network connectivity.

It provides a clear understanding of:

- LAN and WAN integration
- Static routing configuration
- IP addressing scheme
- Router and switch functionality in internetwork communication

This design can be extended further by adding more routers or implementing dynamic routing protocols.